Pierce Lonergan

Software Engineer III · Data & Applied ML

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PROFESSIONAL SUMMARY

Software Engineer III at JPMorganChase specializing in large-scale streaming and batch data infrastructure, with hands-on applied-ML and retrieval work. Designs reusable data platforms on Apache Kafka, Spark, Iceberg, Snowflake, and AWS, and builds Retrieval-Augmented Generation and semantic schema-matching systems using hybrid lexical and dense retrieval with ColBERT reranking. Three promotions in four years. Scientific foundation (B.S. Biochemistry and B.S. Computer Science) with a bias toward measurement and honest evaluation. Open to applied ML, data engineering, and platform roles.

TECHNICAL SKILLS

Streaming & Big Data: Apache Kafka, Apache Spark (Structured Streaming), Apache Iceberg, Apache Avro, Project Reactor, Cassandra, schema evolution, CDC, exactly-once processing

Lakehouse & Warehouse: Snowflake, Iceberg lakehouse, Parquet, partitioning, data modeling, incremental sync

Applied ML & Retrieval: RAG, hybrid retrieval (BM25 + dense), ColBERT / late-interaction reranking, cross-encoder reranking, BGE embeddings, Qdrant, LLM application development, INT8 quantization

Cloud & Infrastructure: AWS (S3, EMR, MSK, Glue, Kinesis, Lambda), Docker, CI/CD, hexagonal architecture, microservices

Data Governance: Canonical catalogs, semantic schema matching, entity resolution, lineage, data-quality validation

Languages: Python, Java, Scala, Groovy, Bash, SQL

PROFESSIONAL EXPERIENCE

JPMorganChase | Consumer & Community Banking, Data Engineering — Columbus, OH

Software Engineer III (Feb 2026 – Present) · Associate Software Engineer (Jan 2024 – Feb 2026) · Software Engineer (Apr 2022 – Jan 2024)

- **Architected reusable streaming-pipeline infrastructure** with declarative source/sink/transform factories, forward-compatible schema evolution, and generic recursive flattening for nested data, lifting the team from roughly **2 to 30 production pipelines per month**.
- Standardized pipeline lifecycle concerns once and reused them across every pipeline: checkpointing, error handling, and graceful shutdown.
- **Shipped data-governance automation that cut weeks from data-approval cycles** and became a production internal product; drove schema-evolution standards, semantic schema matching, and entity resolution against a canonical catalog.
- Built high-performance **Spark ETL** with asynchronous processing and parallelization, and integrated **Kafka** for near-real-time distributed streaming, reducing latency and improving reliability and scalability.
- Modernized big-data systems with reactive paradigms (Project Reactor), pairing Kafka messaging with Cassandra reconciliation and Spark ETL, and migrated high-throughput systems to federated AWS.
- Led a hackathon team to **4th place globally**; architects projected the platform would save roughly 80 engineer-hours per month.

SELECTED PROJECTS

NexusMatcher | Semantic schema-matching system — Personal project · Python

- Enterprise-grade multi-stage retrieval, neural reranking, and learned type projections (hybrid BM25 + dense, ColBERT reranking, Qdrant). **100% Precision@1**, sub-4ms rerank latency, 433 passing tests.

MOMENTUM-X | LLM-driven algorithmic trading system — Personal project · Python

- End-to-end intraday trading system on the Alpaca API (phase-based stop-loss, live daemon, full observability). Designed and rigorously evaluated a multi-agent LLM ensemble, surfacing that ensemble consensus was anti-predictive and redirecting toward a validated execution edge; built mx-arena, a local exchange simulator for backtesting.

NexusPiercer | Nested-schema flattening toolkit — Personal project · Java / Groovy

- Flattens and reconstructs deeply nested JSON and Avro into flat, Spark-ready structures with rich metadata and proven round-trip fidelity.

EDUCATION

The Ohio State University | B.S. Biochemistry and B.S. Computer Science & Engineering — Aug 2016 – Dec 2021

Best Data Visualization, ASA DataFest 2021 · Coursework: machine learning, distributed computing, multithreading, database systems.